MIL-R-12934/20B
6 July 1971
SUPERSEDING
MIL-R-12934/20A
28 January 1971

MILITARY SPECIFICATION

RESISTORS, VARIABLE, WIRE-WOUND, PRECISION, STYLE RR1400

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

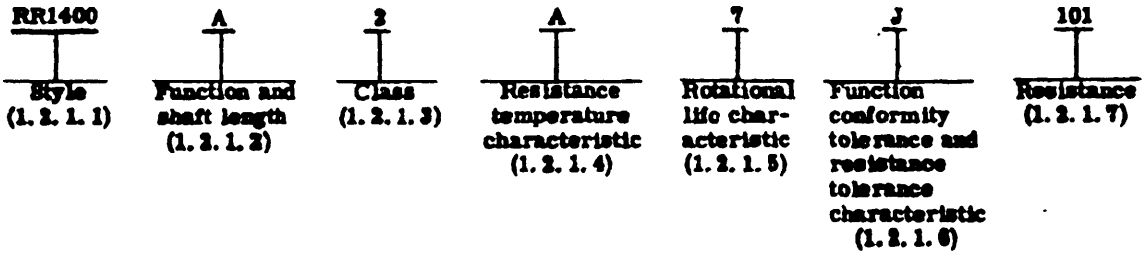
1. SCOPE

1.1 This specification covers the detail requirements for style RR1400, variable, wire-wound, precision resistors.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:

Single-section (cup) resistors



1.2.1.1 Style RR1400. Style RR1400 is applicable to this specification.

1.2.1.2 Function and shaft length. The function and shaft length applicable to this specification shall be identified by a single symbol in accordance with table I.

TABLE I. Function and shaft length.

Symbol	Shaft length Tolerance ±1/32 (.79)	Function
A - - - - -	3/8 (9.53)	Linear
B - - - - -	1/2 (12.70)	Linear
C - - - - -	5/8 (15.88)	Linear
D - - - - -	3/4 (19.05)	Linear
E - - - - -	7/8 (22.23)	Linear
F - - - - -	1 (25.40)	Linear

1.2.1.3 Class. The class applicable to this specification shall be symbols 2 and 3.

1.2.1.4 Resistance temperature characteristic. The resistance temperature characteristics applicable to this specification are symbols A to H, inclusive.

1.2.1.5 Rotational life characteristics. The rotational life characteristics applicable to this specification are symbols 7, 8, and 9. This potentiometer is capable of operation up to 100 RPM.

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1.2.1.6 Function conformity tolerance and resistance tolerance characteristics. The function conformity tolerance and resistance tolerance characteristics applicable to this specification are symbols G to L, inclusive and S to X, inclusive.

1.2.1.7 Resistance. The nominal total resistance values applicable to this specification shall be in accordance with table II.

TABLE II. Nominal total resistance and nominal resolution characteristics.

Nominal total resistance value	Nominal resolution
(Ohms)	(Percent)
200	.040
500	.035
1,000	.03
2,000	.025
5,000	.019
10,000	.016
20,000	.012
50,000	.01
(megohm)	
.1	.008
1.2	.007

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-12934 - Resistors, Variable, Wire-wound, Precision.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-12934 and as specified herein.

3.2 Design and construction. The resistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.3 Minimum resistance. Minimum total resistance value shall be 200 ohms.

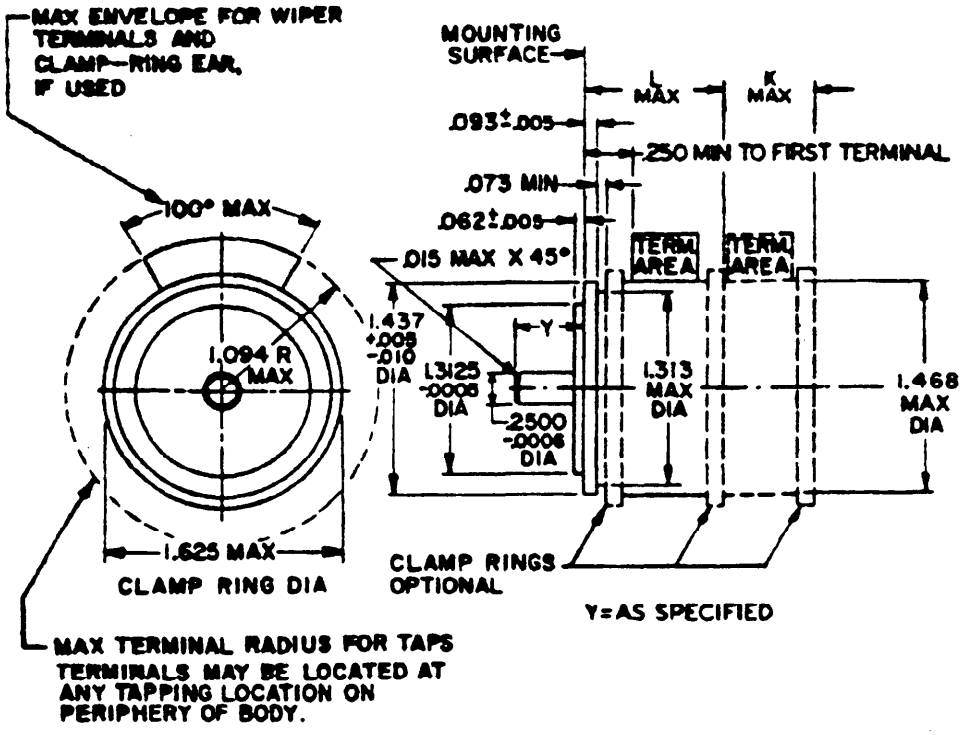
3.4 Maximum resistance. Maximum total resistance value shall be 0.2 megohms.

3.5 Mechanical travel. Mechanical travel shall be $3,600 \, {}^{+10^\circ}_{-0^\circ}$.

3.6 Actual electrical travel. The actual electrical travel shall be $3,600 \, {}^{+4^\circ}_{-0^\circ}$.

3.7 Ganged cups. There shall be no more than 2 cups ganged.

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INCHES	MM	INCHES	MM
.0005	.01	.2500	6.35
.0006	.02	.250	6.35
.005	.13	1.094	27.79
.010	.25	1.3125	33.32
.031	.79	1.313	33.35
.062	1.57	1.437	36.50
.073	1.85	1.468	37.29
.093	2.36	1.625	41.28

Style	Turns	L (max)	K (max)	Maximum continuous working voltage (volts)	Power rating (watts) 1/	Maximum starting and running torque (ounce-inch)				Stop-torque (pound-inch)
						Ten turn, single cup		Per each additional ten-turn cup		
						starting	running	starting	running	
RR1400	Ten	2.250 (57.15)	2.000 (50.80)	500	3.0	1.0	.7	.8	.6	8

1/ When ten turn units are ganged all cups will be 75% of rated wattage.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.

FIGURE 1. Style RR1400.

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3.8 Phasing. For independent linearity the phasing between the cups shall be within ±1°. This shall be measured, with respect to the first cup, starting from the counterclockwise end point.

3.9 Function. The type of function applicable to this specification is linear function (independent linearity).

3.10 Weight. The maximum weight shall be .31 pound and each additional cup shall be .25 pound.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-R-12934 and as specified herein.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-12934.

6. NOTES

6.1 The notes specified in MIL-R-12934 are applicable to this specification.

6.2 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy -
Air Force - 19

Preparing activity:
Army - EL

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